Department of Computer Science and Engineering

Semester: Spring 2026

Course Title: Electrical Circuits

Course Code: CSE209

Section: 06

Name of project:

PSpice Analysis for Maximum Power Transfer

Submitted By:

Group: 2

MD. Tawsif Islam; 2025-1-60-194

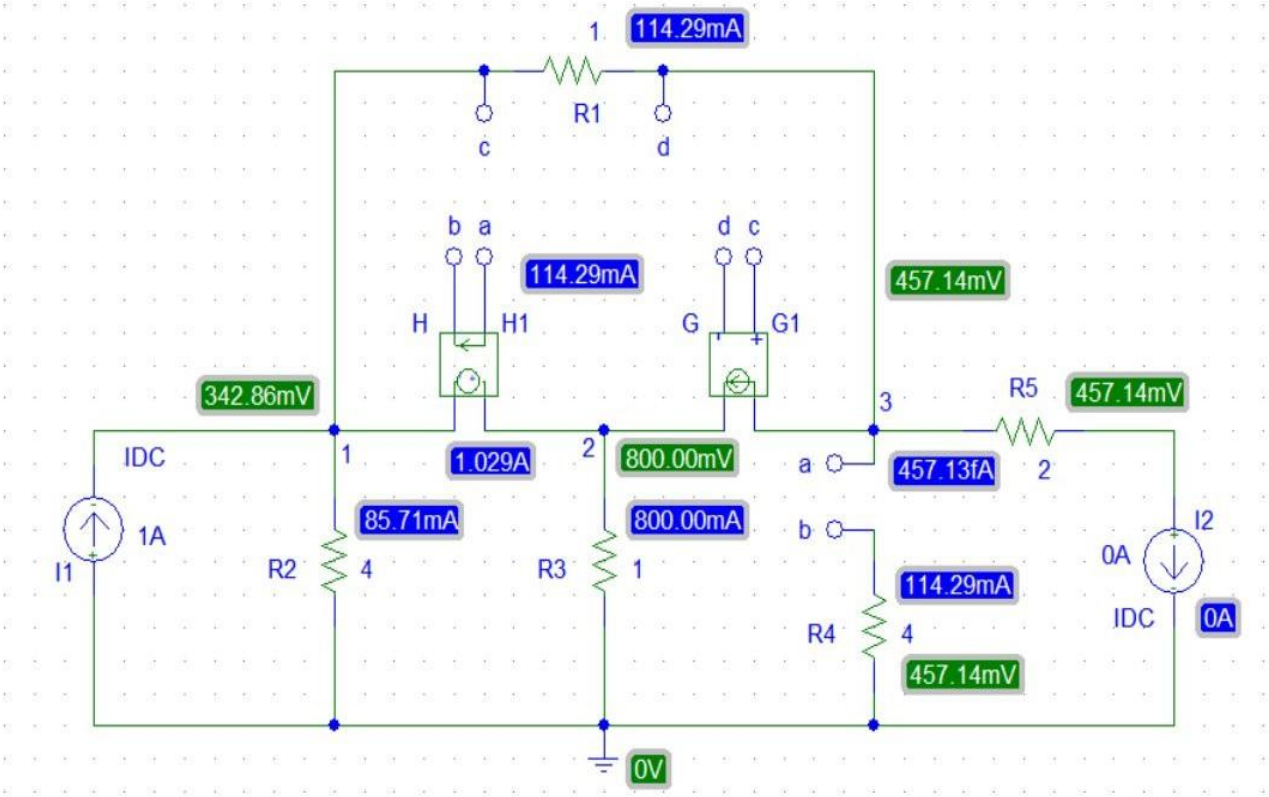
Sajed Bin Moin; 2025-1-60-296

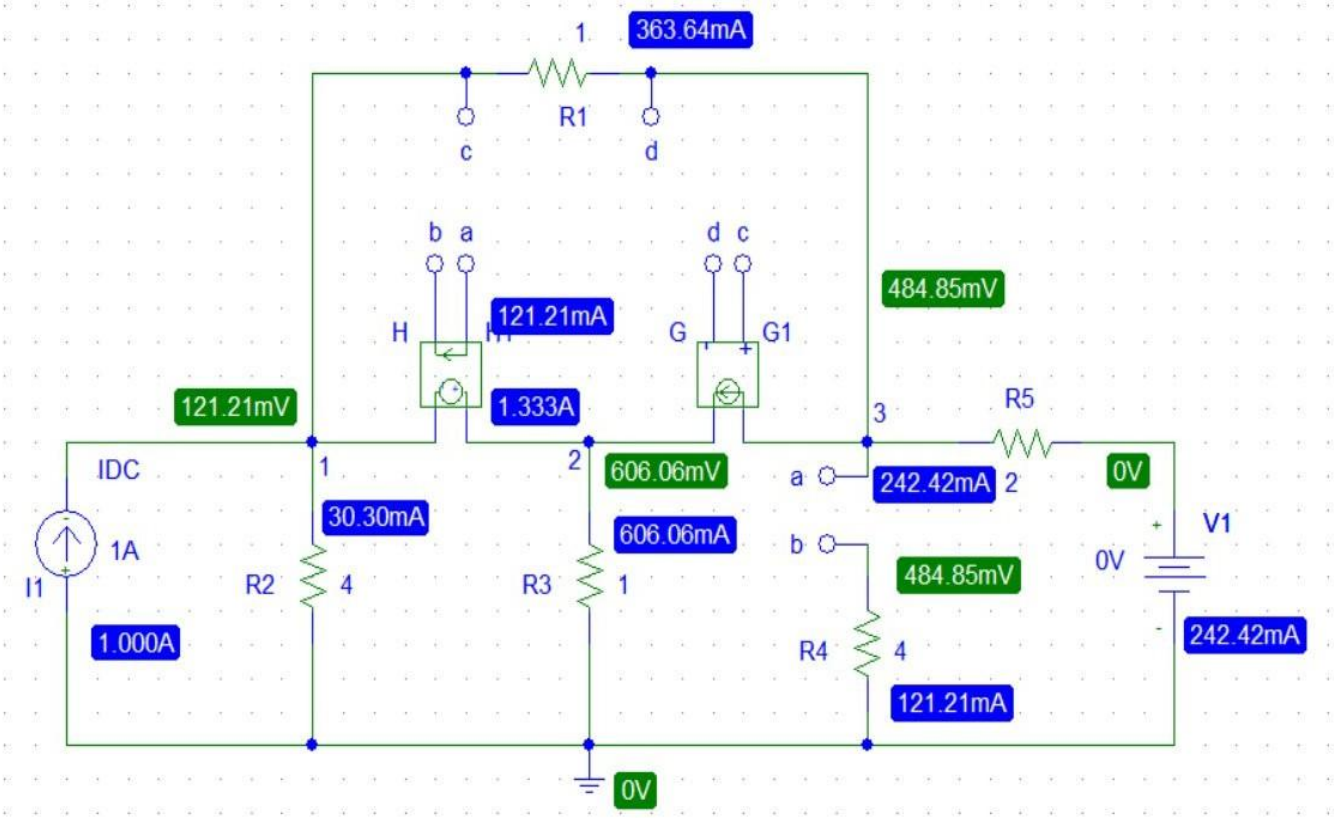
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Submission Date: 28th April 2026

Circuit Diagram :





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From PSpice circuit diagrams,

$V_{th} = V_{oc} = 0.4571V$

$I_{sc} = 0.2424A$

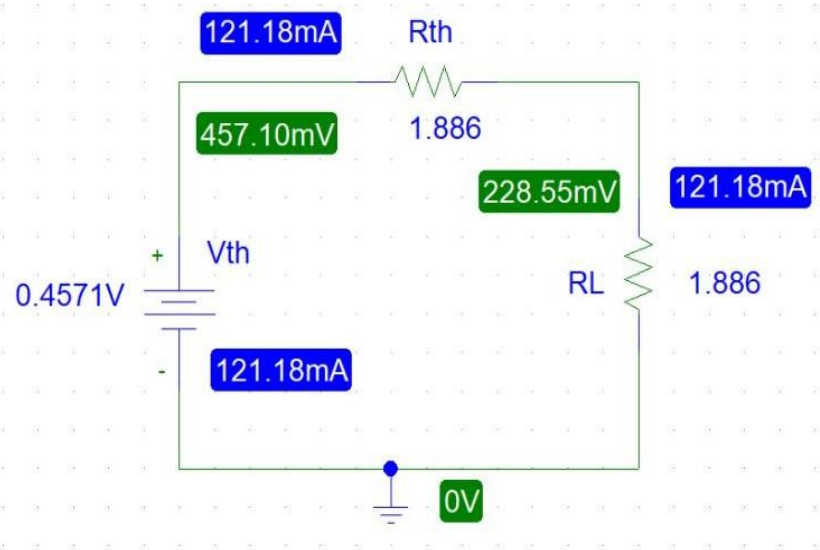
$R_{th} = \frac{V_{oc}}{I_{sc}} = \frac{0.4571}{0.2424} = 1.886\Omega$

For maximum power transfer, $R_L = R_{th}$

$R_L = 1.886\Omega$

$P_{max} = (V_{oc})^2/4*Rth = (0.4571)^2/4*1.886$

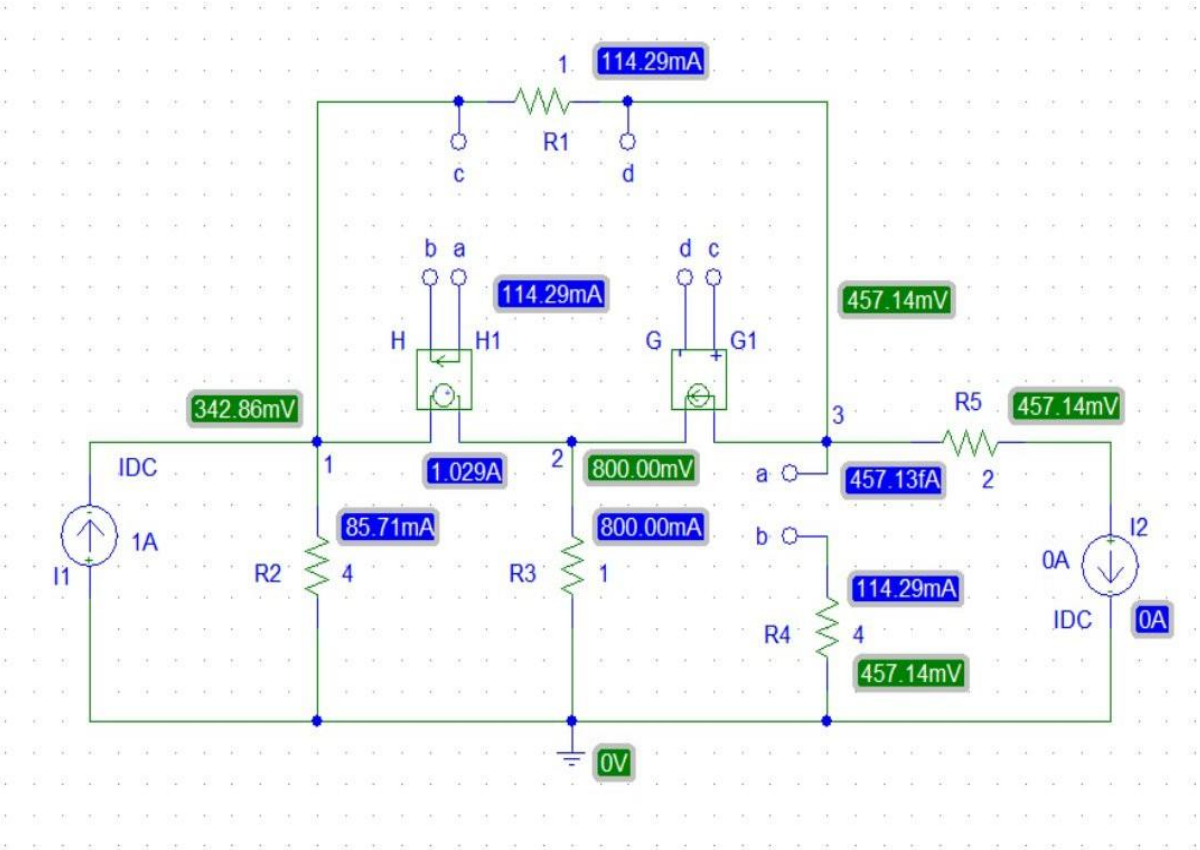
$= 0.02769W = 27.696mw$



Thevenin's equivalent of the circuit

Theoretical solution:

Vth :



Applying KCL at supernode :

-3V1+ 4V2+4V3 = 4 …. (1)

From supernode :

-V1 + V2 – V3 = 0 … (2)

Applying KCL at node V3 :

4V1 – 3V3 = 0 … (3)

Solving (1) , (2) , (3) we get ,

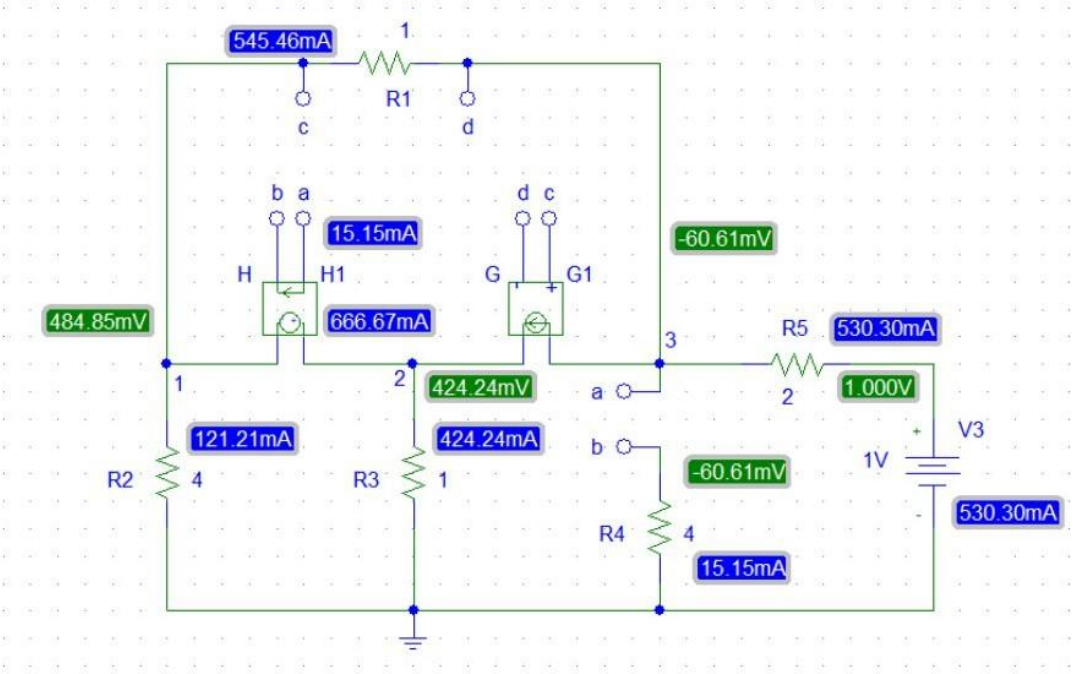
V1 = 0.3428V

V2 = 0.8V

V3 = Voc = 0.4571V

Vth = 0.4571V

R_{th}:



Applying KCL at Supernode:

$-3V_1 + 4V_2 + 4V_3 = 0$ (1)

From Supernode:

$-V_1 + V_2 - V_3 = 0$ (2)

.

Applying KCL at node V3 :

4V1 – V3 = 2 … (3)

Solving (1) , (2) ,(3) we get ,

V1 = 0.485V

V2 = 0.424V

V3 = - 0.060V

Io = 1- V3 / 2 = 1- (- 0.060)/2 = 0.53 A

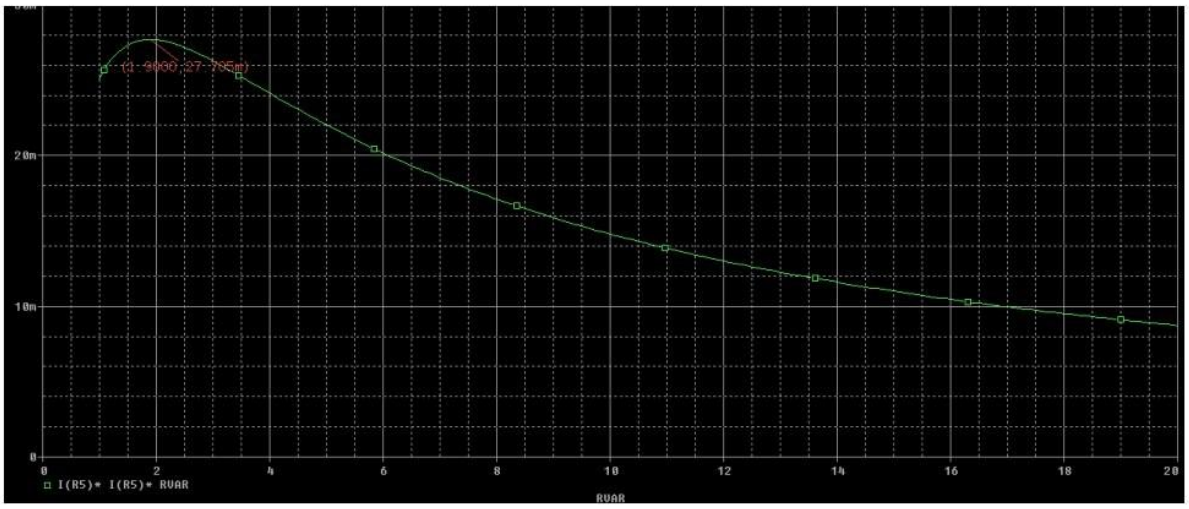
Rth = v0/ i0 = 1/ 0.53 = 1.886ohm

For maximum power transfer , RL = Rth

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RL = 1.886ohm

Pmax = $= (VVVh)\text{^}2/4*\text{Rth} = (0.4571)\text{^}2/4*1.886 = 0.02769\text{W} = 27.696\text{mw}$



From PSpice Simulation with resistance sweep,

$R_L = 1.9\Omega$

$P_{max} = 27.705\text{mW}$

Comparison between the values of R_L and P_{max} obtained in steps 2 and 3.

	Step 2	**Step 3**
R_L	1.886Ω	1.9Ω
P_{max}	27.696mW	27.705mW

Comment: Values of R_L and P_{max} obtained in steps 2 and 3 are pretty much the same.

Conclusion :

The main objective of the project was to analyze the maximum power transfer using PSpice. After comparingthe theoretically obtained values and PSpice measured values of R_L and P_{max}, it is quite visible that both the values of R_L and P_{max} are almost same.

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